Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/19/2005
PI Name:	Sack, Fred Ph.D.		
Project Title:	Development of Gravity Sensitive Plant Cells (Ceratodon) in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	42310	**Congressional District:**	15
Comments:	Deceased August 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	93-OLMSA-01
Start Date:	07/15/1999	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:	Chu, Amy	**Contact Phone:**	650-604-2418
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Flight Program:	Shuttle		
Flight Assignment:	STS-107		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG10-0263		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Little is known about how gravity regulates the distribution of high-density organelles or the direction of plant cell growth. Dark-grown cells of the moss Ceratodon grow away from the gravity vector. These cells probably use amyloplasts that sediment to sense the g- vector. Only a subset of amyloplasts sediment along the length of the cell. If gravity were the sole determinant of plastid distribution and cell orientation, then both would be randomly distributed in microgravity. Instead, both were found to be non-random in microgravity (STS-87, SPM-A). To extend and expand the analysis of the above results, the BRIC-14 experiment is scheduled to fly on STS-107. This study has three objectives: (1) to analyze the cytoskeletal basis of the non-random plastid distribution, (2) to determine the temporal parameters of non-random tip-growth, and (3) to determine whether phototropism occurs in microgravity at a range of low light intensities.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions about how the internal organization and the growth pattern of specialized cells are affected by gravity.
Task Progress:	In addition to shoots and roots, the gravity vector orients the growth of specialized cells such as the apical cell of dark-grown moss protonemata. Each apical cell of the moss Ceratodon purpureus senses the g-vector and adjusts polar growth accordingly producing entire cultures of upright protonemata (negative gravitropism). The effect of withdrawing a constant gravity stimulus on moss growth was studied on two NASA Space Shuttle (STS) missions as well as during clinostat rotation on earth. Cultures grown in microgravity (spaceflight) on the STS-87 mission exhibited two successive phases of non-random growth and patterning, a radial outgrowth followed by the formation of net clockwise spiral growth. Also, cultures pre-aligned by unilateral light developed clockwise hooks during the subsequent dark period. The second spaceflight experiment flew on STS-107 which disintegrated during its descent on February 1, 2003. However, most of the moss experimental hardware was recovered on the ground, and most cultures, which had been chemically fixed during spaceflight, were retrieved. Almost all intact STS-107 cultures displayed strong spiral growth. Non-random culture growth including clockwise spiral growth was also observed after clinostat rotation. Together these data demonstrate the existence of default non-random growth patterns that develop at a population level in microgravity, a response that must normally be overridden and masked by a constant g-vector on earth.
Bibliography Type:	Description: (Last Updated: 05/05/2008)
Abstracts for Journals and Proceedings	VD Kern*, DW Reed, FD Sack "Post accident recovery of hardware and moss cultures from STS-107." COSPAR meeting, Paris, July 18-25, 2004 Abstract #A-00670, , Jul-2004